

SYEED HUSSAIN

B.E. Electronics and Communication Engineering Student

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PROFESSIONAL SUMMARY

Driven Electronics and Communication Engineering student with a strong focus on Embedded Systems, Semiconductor Technology, Core Electronics, and Edge AI. Demonstrated ability to bridge hardware and software through hands-on hackathon experience, technical certifications, and real-world project prototyping. Adept at problem-solving, debugging, and collaborating in high-pressure technical environments.

EDUCATION

B.E. Electronics and Communication Engineering

2025 – 2029

Sri Shakthi Institute of Engineering and Technology, Coimbatore

- **First-Year CGPA:** 8.0/10
- **Core Coursework:** Digital Electronics, Electronic Circuits, Semiconductor Devices, Microprocessors & Microcontrollers, HDL/Verilog

TECHNICAL SKILLS

Programming Languages	Python, C, C++
Hardware & Embedded	Arduino, ESP32, Digital Electronics, Electronic Circuits, Semiconductor Fundamentals, HDL (Basic)
AI & Computer Vision	OpenCV, YOLOv8, Edge AI, TinyML, Neural Network Optimization
Web Tech & Tools	React, Vite, Flask, VS Code, Git, GitHub, Figma
Languages	English, Tamil, Hindi, Urdu

HACKATHON & PROJECT EXPERIENCE

HACKSPRINT'26 (24-Hour AI Hackathon)

July 2026

AI Research Laboratory, Sri Shakthi Institute — BrainBoard AI (Computer Vision Developer)

- Engineered an AI-based CCTV hostel security monitoring concept to enhance physical security operations.
- Developed computer vision pipelines using Python, OpenCV, and YOLOv8 for real-time area detection.
- Implemented automated tracking algorithms for loitering and overcrowding detection.
- Led the technical integration of the AI/Computer Vision module and delivered the final project demonstration to judges.

VLSITHACK'26 (24-Hour VLSI Hackathon)

2026

Department of EEE (VLSI Design and Tech) — SparONIU (Hardware & AI Optimization Developer)

- Explored and implemented neural-network optimization strategies specifically designed for edge/NPU-oriented inference.
- Optimized computational efficiency using MLP weight pruning, model sparsity techniques, and Run-Length Encoding (RLE).
- Bridged the gap between theoretical AI algorithms and hardware-efficient computing applications.

SIH Internal Hackathon (Student Innovation)

2026

Travel & Tourism Theme — Software & Network Prototype Developer

- Designed a software-based cellular network monitoring concept to optimize traveler connectivity.
- Built a highly responsive prototype dashboard utilizing React and Vite.
- Programmed data visualizations for 4G/5G network selection, signal strength, download/upload speeds, latency, and overall connection quality.
- Delivered the technical presentation, effectively communicating complex network metrics to non-technical audiences.

CERTIFICATIONS

- **Python Course for Beginners:** Mastering the Essentials – Scaler (2026)
- **Free Embedded Systems Course:** Setup – BrainAcademy (2026)
- **Embedded for Beginners:** Circuit Design – NILT (2026)